Análisis del circuito Por Superposicion

Circuito 1 : APagando $F_0(t)$

$$X(t) = X_1(t) + X_2(t)$$

$$X(t) =$$

$$X_1(t) = C_s \frac{d[F(t - F_s(t)]}{dt}$$

$$X_2(t) = \frac{F(t) - F_{s1}(t)}{t}$$

$$C_p = \frac{dF_{s1}(t)}{dt} = C_s \frac{d[F(t) - F_{s1}(t)]}{dt} + \frac{F(t) - F_{s1}(t)}{R}$$

$$C_p s F_{s1}(s) = C_s s [F(s) - F_{s1}(s)] + F(s) - \frac{F_{s1}(s)}{R}$$

$$C_p s F_{s1}(s) = C_s s F(s) - C_s s F_{s1}(s) + \frac{F(s)}{R} - \frac{F_{s1}(s)}{R}$$

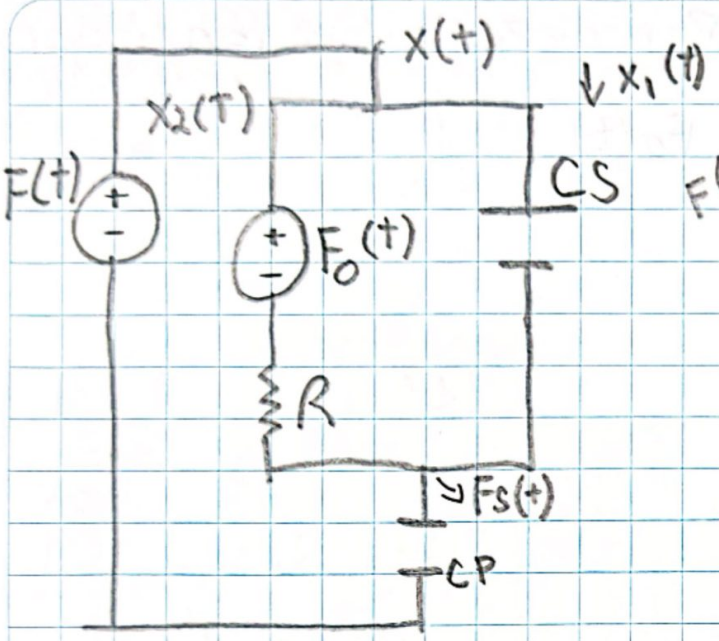
$$\left(C_p s + C_s s + \frac{1}{R}\right) F_s(s) = \left(C_s s + \frac{1}{R}\right) F(s)$$

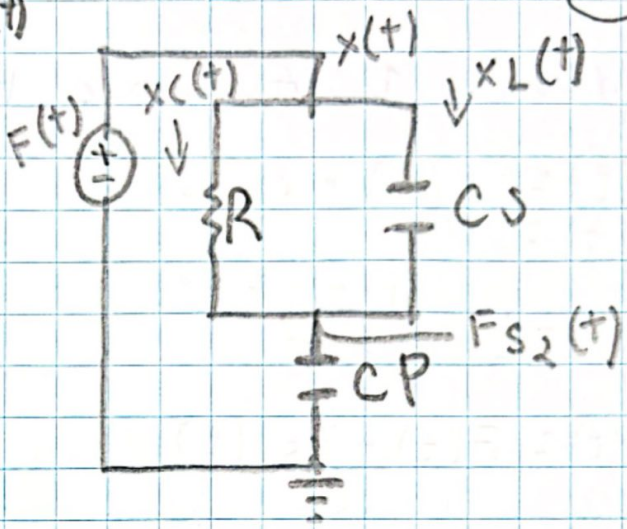
$$\frac{R(C_p + C_s)s + 1}{R} F_{s1}(s) = \frac{C_s R s + 1}{R} F(s)$$

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Circuito 2: Apagando F(t)

$$-\alpha F(t) = R\,x(t) + \frac{1}{C_P + C_S}\int x(t)\,dt$$

$$F_{S2}(t) = \frac{1}{C_P + C_S}\int x(t)\,dt$$

$$-\alpha F(s) = R\,X(s) + \frac{X(s)}{(C_P + C_S)s}$$

$$F_{S2}(s) = \frac{X(s)}{(C_P + C_S)s}$$

$$F(s) = \frac{X(s)}{\alpha}\,\frac{R(C_P + C_S)s + 1}{(C_P + C_S)s}$$

$$F_{S2}(s) = -\frac{\alpha F(s)}{R(C_P + C_S)s + 1}$$

Función de transferencia

$$F_S(s) = F_{S1}(s) + F_{S2}(s) = \frac{C_S R_S + 1}{R[C_P + C_S]s + 1}F(s) - \frac{\alpha F(0)}{R[C_P + C_S]s + 1}$$

$$F_S(s) = \frac{C_S R_S + 1 - \alpha}{R[C_P + C_S]s + 2}F(s) \rightarrow \text{salida final}$$

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